

XPath Logics

(for configuring Interaction Alerts):

Objective:

- Given below are the XPath Expressions that can be used to run below checks on the Result.xml to
- Find if there are any interaction alerts for any "specific drug" or "severity" or "irrespective of any drug or severity" in the response.xml
 - These flags can be used to pop-up interaction alert or over-ride reasoning window in your application only when this XPATH Flag is "True"
 - XPATH Flags
 - if XPATH Count > 0 = True
 - if XPATH Count = 0 = False

Example or one-use case:

If a specific drug being prescribed has any interaction with other drugs prescribed in same order or with previously prescribed Medications which are still active, then use the below XPath logic to run that check before applying the stylesheet. Hence if the XPath expression reports count > 0 then the Result.xml output has interaction for the specific medicine being prescribed, only then you proceed triggering pop-up window to display interaction alert or same flag can be used to trigger over-ride reasoning window via your application else not.

Applicable Modules:

These logics cover "Drug to Drug", "Drug to Allergy", "Drug to Health", "Duplicate Therapy", "Duplicate Ingredient", "Drug Pregnancy", "Drug Lactation", "Drug Dose" Interaction Alerts

a) Displaying Interaction Alert only if the interaction alert is available in the Result.xml output

If the specific drug being prescribed has any interaction with other drugs prescribed in same order or with previously prescribed Medications which are still active, then use the below XPath logic to run that check before applying the stylesheet. Hence if the XPath expression reports count > 0 then the Result.xml output has interaction for the specific medicine being prescribed, then you can apply the stylesheet to display the interaction on the application UI

These logics cover "Drug to Drug", "Drug to Allergy" and "Drug to Health" Interactions.

(i) XPath for "Drug to Drug Interaction":

To check if any Drug to Drug Interaction alert exists in the Result.xml

For Specific Drug whose GUID = {864EBD9F-9789-4451-908A-C6A354E9FA24}

```
"count(//Interaction/*[@reference='{864EBD9F-9789-4451-908A-C6A354E9FA24}' and  
not(@Mirror)]/Route/*/Route/ClassInteraction|//Interaction/*/Route/*[@reference='{864EBD9F-  
9789-4451-908A-C6A354E9FA24}' and not(@Mirror)]/Route/ClassInteraction)"
```

For Specific Drug whose GUID = {864EBD9F-9789-4451-908A-C6A354E9FA24} and Severity

```
"count(//Interaction/*[@reference='{864EBD9F-9789-4451-908A-C6A354E9FA24}' and  
not(@Mirror)]/Route/*/Route/ClassInteraction/Severity[@name='Severe' or @name='Moderate' or  
@name='Minor' or @name='Caution']|//Interaction/*/Route/*[@reference='{864EBD9F-9789-  
4451-908A-C6A354E9FA24}' and not(@Mirror)]/Route/ClassInteraction/Severity[@name='Severe' or  
@name='Moderate' or @name='Minor' or @name='Caution'])"
```

Irrespective to any specific drug

```
"count(//Interaction/*[not(@Mirror)]/Route/*/Route/ClassInteraction|  
//Interaction/*/Route/*[not(@Mirror)]/Route/ClassInteraction)"
```

Irrespective to any specific drug but with specific Severity

```
"count(//Interaction/*[not(@Mirror)]/Route/*/Route/ClassInteraction/Severity[@name='Severe' or  
@name='Moderate' or @name='Minor' or  
@name='Caution']|//Interaction/*/Route/*[not(@Mirror)]/Route/ClassInteraction/Severity[@name  
='Severe' or @name='Moderate' or @name='Minor' or @name='Caution'])"
```

(ii) XPath for "Drug to Health Interaction":

To check if any Drug to Health Interaction alert exists in the Result.xml

For Specific Drug whose GUID = {35051B69-F816-49D2-8A1A-1DA953855A0E}

```
"count(//Interaction/*[@reference='{35051B69-F816-49D2-8A1A-1DA953855A0E}'and  
not(@Mirror)]/Route/HealthIssueCode/ClassInteraction|//Interaction/*/Route/*[@reference='{350  
51B69-F816-49D2-8A1A-1DA953855A0E}' and  
not(@Mirror)]/Route/HealthIssueCode/ClassInteraction)"
```

For Specific Drug whose GUID = {35051B69-F816-49D2-8A1A-1DA953855A0E} and Severity

```
"count(//Interaction/*[@reference='{35051B69-F816-49D2-8A1A-1DA953855A0E}'and  
not(@Mirror)]/Route/HealthIssueCode/ClassInteraction/Severity[@name='Contraindicated' or  
@name='Extreme Caution']|//Interaction/*/Route/*[@reference='{35051B69-F816-49D2-8A1A-  
1DA953855A0E}' and  
not(@Mirror)]/Route/HealthIssueCode/ClassInteraction/Severity[@name='Contraindicated' or  
@name='Extreme Caution'])"
```

Irrespective to any specific drug

```
"count(//Interaction/*[not(@Mirror)]/Route/HealthIssueCode/ClassInteraction|//Interaction/*/Rou  
te/*[not(@Mirror)]/Route/HealthIssueCode/ClassInteraction)"
```

Irrespective to any specific drug but with specific Severity

```
"count(//Interaction/*[not(@Mirror)]/Route/HealthIssueCode/ClassInteraction/  
Severity[@name='Contraindicated' or @name='Extreme  
Caution']|//Interaction/*/Route/*[not(@Mirror)]/Route/HealthIssueCode/ClassInteraction/  
Severity[@name='Contraindicated' or @name='Extreme Caution'])"
```

(iii) XPath for "Drug to Allergy Interaction":

To check if any Drug to Allergy Interaction alert exists in the Result.xml

For Specific Drug whose GUID = {673FE2DD-4B75-4EFA-85DA-7957A90AF7A2}

```
"count(//Interaction/*[@reference='{673FE2DD-4B75-4EFA-85DA-7957A90AF7A2}' and not(@Mirror)]/Allergy)"
```

For any Drug to Allergy Interactions, irrespective to any specific drug

```
"count(//Interaction/*[not(@Mirror)]/Allergy)"
```

(iv) XPath for "Duplicate Ingredient Alert":

To check if any Duplicate Ingredient alert exists in the Result.xml

For Specific Drug whose GUID = {B3E5E9FF-0857-4B5E-E034-080020E1DD8C}

```
"count(//Interaction/DuplicateIngredient/Warning/Molecule/Molecule/*[@reference='{B3E5E9FF-0857-4B5E-E034-080020E1DD8C}'])"
```

For Specific Drug whose GUID = {B3E5E9FF-0857-4B5E-E034-080020E1DD8C} and Severity

```
"count(//Interaction/DuplicateIngredient/Warning[@Level='1' or @Level='2' or @Level='3']/Molecule/Molecule/*[@reference='{B3E5E9FF-0857-4B5E-E034-080020E1DD8C}'])"
```

Irrespective to any specific drug

```
"count(//Interaction/DuplicateIngredient/Warning/Molecule/Molecule)"
```

Irrespective to any specific drug but with specific Severity

```
"count(//Interaction/DuplicateIngredient/Warning[@Level='1' or @Level='2' or @Level='3']/Molecule/Molecule)"
```

(v) XPath for "Duplicate Therapy Alert":

To check if any Duplicate Therapy alert exists in the Result.xml

For Specific Drug whose GUID = {B4268321-1B08-73AE-E034-080020E1DD8C}

```
"count(//Interaction/DuplicateTherapy/Warning/Duplicate/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}'])"
```

For Specific Drug whose GUID = {B4268321-1B08-73AE-E034-080020E1DD8C} and Severity

```
"count(//Interaction/DuplicateTherapy/Warning[@Level='1' or @Level='2' or @Level='3']/Duplicate/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}'])"
```

Irrespective to any specific drug

```
"count(//Interaction/DuplicateTherapy/Warning/Duplicate)"
```

Irrespective to any specific drug but with specific Severity

```
"count(//Interaction/DuplicateTherapy/Warning[@Level='1' or @Level='2' or @Level='3']/Duplicate)"
```

(vi) XPath for "Drug to Pregnancy Alert":

To check if any Drug to Pregnancy alert exists in the Result.xml

For Specific Drug whose GUID = {707BFAFF-1B76-4621-9983-D8212D2E00F7}

```
"count(//Interaction/*[@reference='{707BFAFF-1B76-4621-9983-D8212D2E00F7}' and not(@Mirror)]/Route/Pregnancy|//Interaction/*/Route/*[@reference='{707BFAFF-1B76-4621-9983-D8212D2E00F7}' and not(@Mirror)]/Route/Pregnancy)"
```

For Specific Drug whose GUID = {707BFAFF-1B76-4621-9983-D8212D2E00F7} and Severity

```
"count(//Interaction/*[@reference='{707BFAFF-1B76-4621-9983-D8212D2E00F7}' and not(@Mirror)]/Route/Pregnancy/Category[@name='A' or @name='B' or @name='C' or @name='D' or @name='X' or @name='+']|//Interaction/*/Route/*[@reference='{707BFAFF-1B76-4621-9983-D8212D2E00F7}' and not(@Mirror)]/Route/Pregnancy/Category[@name='A' or @name='B' or @name='C' or @name='D' or @name='X' or @name='+'])"
```

Irrespective to any specific drug

```
"count(//Interaction/*[not(@Mirror)]/Route/Pregnancy|//Interaction/*/Route/*[not(@Mirror)]/Route/Pregnancy)"
```

Irrespective to any specific drug but with specific Severity

```
"count(//Interaction/*[not(@Mirror)]/Route/Pregnancy/Category[@name='A' or @name='B' or @name='C' or @name='D' or @name='X' or @name='+']|//Interaction/*/Route/*[not(@Mirror)]/Route/Pregnancy/Category[@name='A' or @name='B' or @name='C' or @name='D' or @name='X' or @name='+'])"
```

(vii) XPath for "Drug to Pregnancy Alert" (Women of Childbearing Age [WOBCA]):

To check if any WOCBA alert exists in the Result.xml

For Specific Drug whose GUID = {B4268321-1B08-73AE-E034-080020E1DD8C}

```
"count(//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and not(@Mirror)]/Route/WOCBA|//Interaction/*/Route/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and not(@Mirror)]/Route/WOCBA)"
```

For Specific Drug whose GUID = {B4268321-1B08-73AE-E034-080020E1DD8C} and Severity

```
"count(//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and not(@Mirror)]/Route/WOCBA/Category[@name='A' or @name='B' or @name='C' or @name='D' or @name='X' or @name='+']|//Interaction/*/Route/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and not(@Mirror)]/Route/WOCBA/Category[@name='A' or @name='B' or @name='C' or @name='D' or @name='X' or @name='+']]"
```

Irrespective to any specific drug

```
"count(//Interaction/*[not(@Mirror)]/Route/WOCBA|//Interaction/*/Route/*[not(@Mirror)]/Route/WOCBA)"
```

Irrespective to any specific drug but with specific Severity

```
"count(//Interaction/*[not(@Mirror)]/Route/WOCBA/Category[@name='A' or @name='B' or @name='C' or @name='D' or @name='X' or @name='+']|//Interaction/*/Route/*[not(@Mirror)]/Route/WOCBA/Category[@name='A' or @name='B' or @name='C' or @name='D' or @name='X' or @name='+']]"
```

(viii) XPath for "Drug to Lactation Alert":

To check if any Drug to Lactation alert exists in the Result.xml

For Specific Drug whose GUID = {B4268321-1B08-73AE-E034-080020E1DD8C}

```
"count(//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and not(@Mirror)]/Route/Lactation|//Interaction/*/Route/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and not(@Mirror)]/Route/Lactation)"
```

For Specific Drug whose GUID = {B4268321-1B08-73AE-E034-080020E1DD8C} and Severity

```
"count(//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and not(@Mirror)]/Route/Lactation/Severity[@name='Contraindicated' or @name='Avoid if possible' or @name='Caution']|//Interaction/*/Route/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and not(@Mirror)]/Route/Lactation/Severity[@name='Contraindicated' or @name='Avoid if possible' or @name='Caution']]"
```

Irrespective to any specific drug

```
"count(//Interaction/*[not(@Mirror)]/Route/Lactation|//Interaction/*/Route/*[not(@Mirror)]/Route/Lactation)"
```

Irrespective to any specific drug but with specific Severity

```
"count(//Interaction/*[not(@Mirror)]/Route/Lactation/Severity[@name='Contraindicated' or @name='Avoid if possible' or @name='Caution']|//Interaction/*/Route/*[not(@Mirror)]/Route/Lactation/Severity[@name='Contraindicated' or @name='Avoid if possible' or @name='Caution'])"
```

(ix) XPath for "Drug Dose Alert":

a) Dose Check

To check if any Drug Dose alert exists in the Result.xml

For Specific Drug whose GUID = {B4268321-1B08-73AE-E034-080020E1DD8C}

```
"count(//Interaction/DoseCheck/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}']/Warnings)"
```

Irrespective to any specific drug

```
"count(//Interaction/DoseCheck/*/Warnings)"
```

For Specific Drug whose GUID = {B4268321-1B08-73AE-E034-080020E1DD8C} and Specific WarningID

```
count(//Interaction/DoseCheck/*[@reference='{A8C10CD4-E8E0-4E54-863C-80AF913F0386}']/Warnings/Warning[@id='IDS_DOSE_MAX_WARNING'])
```

b) Combined Dose Alert

To check if any Combined Drug Dose alert exists in the Result.xml

For Specific Drug whose GUID = {B94A25B3-52FC-49DB-8E5D-629ABC01ECB4}

```
"count(//Interaction/DoseCheck/CombinedDoseCheck/Molecule/*[@reference='{B94A25B3-52FC-49DB-8E5D-629ABC01ECB4}'])"
```

Irrespective to any specific drug

```
"count(//Interaction/DoseCheck/CombinedDoseCheck/Molecule/Warnings)"
```

For Specific Drug whose GUID = {B4268321-1B08-73AE-E034-080020E1DD8C} and Specific WarningID

```
"count(//Interaction/DoseCheck/CombinedDoseCheck/Molecule/*[@reference='{B94A25B3-52FC-49DB-8E5D-629ABC01ECB4}']|//Interaction/DoseCheck/CombinedDoseCheck/Molecule/Warnings/Warning[@id='IDS_DOSE_MAX_WARNING'])"
```

b) Use Concatenated XPath Query for all above modules in application Code:

XPath with GUID: when using each drug (for Instant alert)

```
count(//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/*/Route/ClassInteraction|//Interaction/*/Route/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/ClassInteraction|//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/HealthIssueCode/ClassInteraction|//Interaction/*/Route/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/HealthIssueCode/ClassInteraction|//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Allergy|//Interaction/DuplicateIngredient/Warning/Molecule/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}']|//Interaction/DuplicateTherapy/Warning/Duplicate/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}']|//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/Pregnancy|//Interaction/*/Route/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/Pregnancy|//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/WOCBA|//Interaction/*/Route/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/WOCBA|//Interaction/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/Lactation|//Interaction/*/Route/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}' and
not(@Mirror)]/Route/Lactation|//Interaction/DoseCheck/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}']/Warnings|//Interaction/DoseCheck/CombinedDoseCheck/Molecule/*[@reference='{B4268321-1B08-73AE-E034-080020E1DD8C}']])
```


XPath without GUID:

```
count(//Interaction/*[not(@Mirror)]/Route/*/Route/ClassInteraction|//Interaction/*/Route/*[not(@Mirror)]/Route/ClassInteraction|//Interaction/*[not(@Mirror)]/Route/HealthIssueCode/ClassInteraction|//Interaction/*[not(@Mirror)]/Route/HealthIssueCode/ClassInteraction|//Interaction/*[not(@Mirror)]/Allergy|//Interaction/DuplicateIngredient/Warning/Molecule|//Interaction/DuplicateTherapy/Warning/Duplicate|//Interaction/*[not(@Mirror)]/Route/Pregnancy|//Interaction/*/Route/*[not(@Mirror)]/Route/Pregnancy|//Interaction/*[not(@Mirror)]/Route/WOCBA|//Interaction/*/Route/*[not(@Mirror)]/Route/WOCBA|//Interaction/*[not(@Mirror)]/Route/Lactation|//Interaction/*/Route/*[not(@Mirror)]/Route/Lactation|//Interaction/DoseCheck/*/Warnings|//Interaction/DoseCheck/CombinedDoseCheck/Molecule/Warnings)
```

XPath Query Example for Over-Ride reasoning window:

```
count(//Interaction/*[not(@Mirror)]/Route/*/Route/ClassInteraction/Severity[@name='Severe' or @name='Moderate']|//Interaction/*/Route/*[not(@Mirror)]/Route/ClassInteraction/Severity[@name='Severe' or @name='Moderate']|//Interaction/*[not(@Mirror)]/Route/HealthIssueCode/ClassInteraction/Severity[@name='Contraindicated']|//Interaction/*/Route/*[not(@Mirror)]/Route/HealthIssueCode/ClassInteraction/Severity[@name='Contraindicated']|//Interaction/*[not(@Mirror)]/Allergy|//Interaction/DuplicateIngredient/Warning[@Level='1' or @Level='2']/Molecule|//Interaction/DuplicateTherapy/Warning[@Level='1' or @Level='2']/Duplicate|//Interaction/*[not(@Mirror)]/Route/Pregnancy/Category[@name='D' or @name='X' or @name='+']|//Interaction/*/Route/*[not(@Mirror)]/Route/Pregnancy/Category[@name='D' or @name='X' or @name='+']|//Interaction/*[not(@Mirror)]/Route/WOCBA/Category[@name='D' or @name='X' or @name='+']|//Interaction/*/Route/*[not(@Mirror)]/Route/WOCBA/Category[@name='D' or @name='X' or @name='+']|//Interaction/*[not(@Mirror)]/Route/Lactation/Severity[@name='Contraindicated' or @name='Avoid if possible']|//Interaction/*/Route/*[not(@Mirror)]/Route/Lactation/Severity[@name='Contraindicated' or @name='Avoid if possible']|//Interaction/DoseCheck/*/Warnings/Warning[@id='IDS_DOSE_MAX_WARNING']|//Interaction/DoseCheck/CombinedDoseCheck/Molecule/Warnings/Warning[@id='IDS_DOSE_MAX_WARNING'])
```